

HTAD-06: AI + IoT for Early Detection of Anemia and Malnutrition

Team Briefing Document — Problem Statement, Build Plan & Role Split

1. The Problem Statement (HTAD-06)

Official brief: Develop a mobile diagnostic device using IoT sensors and AI to detect anemia and malnutrition in children and pregnant women non-invasively.

SDG 2 SDG 3 SDG 10

Why we picked this one: Anemia and malnutrition screening today needs a blood draw and a lab, which is exactly why most at-risk women and children in rural/underserved areas never get screened. A non-invasive, sensor-based device that flags risk on the spot is both a real hard-tech build and a problem judges immediately understand the impact of. Most other teams at this hackathon will not have the embedded systems background to pull off a real sensor pipeline in the time given — this is where we have a genuine edge.

2. What We Are Building

A handheld, fingertip device that reads a person's pulse waveform through their finger (no needles, no blood), and uses that signal to estimate whether they are at risk of anemia. The device shows the result instantly on a small screen and lights up Green / Yellow / Red depending on risk level, and can also send the reading to a simple dashboard for a health worker to log and track over time.

How it actually works, step by step:

- **Sense:** A small sensor (MAX30102) shines a light through the fingertip and measures how the blood absorbs it. This is the same core tech used in pulse oximeters.
- **Read:** The ESP32 (a small WiFi-enabled microcontroller — think 'tiny computer') continuously reads this signal for about 15-30 seconds.
- **Extract:** From that raw signal we calculate three numbers: SpO2 (oxygen level), heart rate variability, and Perfusion Index (a measure of blood flow strength — this is the number most strongly linked to anemia in published research).
- **Classify:** These numbers are fed into a simple risk model — partly rule-based thresholds (backed by published medical correlations) and partly a small trained classifier if time allows — to output a Low / Medium / High anemia risk.
- **Display:** The result shows instantly on a small OLED screen on the device itself, and a Red/Yellow/Green LED lights up so it's understandable at a glance, even by someone with no medical training.
- **Log (stretch):** The reading can also be sent securely over WiFi to a simple web dashboard, so a health worker can track a patient's readings over multiple visits.

Our edge: we're layering in basic data security on the WiFi transmission (encrypting the reading before it leaves the device) — drawing directly from the IAM/HSM work some of us have already done. Almost no other team will think to secure patient health data at a hackathon, and it's a strong answer if judges ask 'what happens to this data.'

3. Hardware We're Using

Component	What it does	Qty
ESP32 DevKit v1	The 'brain' — a small WiFi-capable microcontroller that runs our code	1
MAX30102 sensor	Reads pulse signal from the fingertip (SpO2, heart rate, perfusion index)	1
0.96" OLED display	Shows the live reading and result on the device itself	1
Red / Yellow / Green LEDs	Instant visual risk indicator	3
220 ohm resistors	Protect the LEDs from too much current	3
Push button	Triggers a new reading	1
Breadboard + jumper wires	Connects everything together, no soldering needed for Demo 1	1 set

Everything connects via a breadboard (no soldering) for the first working demo — soldering onto a compact board is a stretch goal only if time allows later.

4. Rough Timeline

Hours 0-4	Get the sensor reading data reliably on the ESP32
Hours 4-10	Calculate SpO2 / HRV / Perfusion Index, wire up screen + LEDs
Hours 10-16	Build the risk classification logic + secure data transmission
Hours 16-22	Build the dashboard, tidy up the physical device
Hours 22-28	Test everything together, prep the pitch, record a backup demo video
Remaining time	Rest in shifts, rehearse the pitch

5. Role Split — Who Does What

Roles are split by strength, but this is a hardware-software integration project, so everyone should understand the full pipeline well enough to explain it to judges, not just their own part.

Person 1 Hardware & Firmware (Sensing side)

- Wire up the ESP32, MAX30102 sensor, OLED, LEDs and button on the breadboard
- Write the firmware that reads the raw pulse signal from the MAX30102
- Extract SpO2 and heart-rate-variability from the raw signal on-device
- Own the physical device stability — make sure readings are consistent for the live demo

Person 2 Hardware & Firmware (Output side) + Security

- Wire up and code the OLED display output and Red/Yellow/Green LED risk indicator logic
- Calculate Perfusion Index from the sensor data and feed it into the risk logic
- Build the secure data transmission from the ESP32 to the dashboard (encryption, basic auth)
- Own the security/data-pipeline pitch point — this is our differentiator vs other teams

Person 3 Dashboard & Frontend Lead

- Build the web dashboard that receives and displays readings over time per patient
- Design the on-device OLED screen UI (what shows during/after a reading)
- Make the demo visually clean — this is what judges see, so polish matters here
- Prepare visuals/slides for the pitch deck

Person 4 ML / Classification & Pitch Lead

- Source a small public dataset (SpO2 / perfusion index vs anemia) to validate our thresholds and, if time allows, train a lightweight classifier
- Own the rule-based fallback logic so the demo never fails even if the trained model isn't ready
- Write and lead the pitch narrative — the SDG story, the technical explanation, and the security angle
- Coordinate the team's timeline and keep the demo scope realistic under time pressure

Golden rule for the pitch: whoever is talking to judges should be able to explain the full pipeline end-to-end, not just their own piece — judges often ask a random team member a question outside their 'assigned' area to test if it's a real team effort.